Secure Elder-Care Monitoring Platform
Video Based

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Elderly individuals receiving home-based care often depend on hired caregivers to perform critical daily tasks such as waking, feeding, bathing, and medication administration. While modern AI and computer vision systems can recognize human activities and detect falls from video streams, most existing solutions do not verify whether these caregiving actions are performed according to explicit, time-bound schedules defined by families or clinicians. As a result, missed or delayed tasks and unauthorized care interactions may go unnoticed.

This project proposes an Activity Schedule Monitoring and Rule-Based Decision Engine as an integral component of a Secure Elder-Care Monitoring Platform. Administrators define daily care schedules (tasks, time windows, responsible caregiver) for each elderly individual, while the main AI subsystem uses pre-trained pose-based Human Activity Recognition (HAR) and face-recognition models from the literature to convert camera streams into discrete activity and identity events. In particular, skeleton- or pose-based HAR models operate on sequences of extracted human skeleton joints to recognize activities (e.g., eating, walking, bathing, sitting, falls), and face-recognition models identify which elder or caregiver is present, without any model training performed inside this component. The schedule monitoring engine then consumes these recognized activity and identity events and evaluates them against the defined schedules: when the recognized activity aligns with the scheduled task, within the allowed time window and by an authorized caregiver, the system automatically marks the task as completed and stores it in the task history. If a required activity is not detected within the specified time window, or if it is performed outside the scheduled window or by an unauthorized person, the module generates a deviation event and immediately notifies the administrator through the platform.

The solution will be implemented using a Python-based backend and a relational database for schedule and task-status storage, integrating with the main platform via REST APIs. Evaluation will be conducted using scenario-based test data and simulated event streams derived from pre-trained Activity Recognition Models, measuring task detection accuracy and correctness of schedule-compliance classification. The expected outcome is a working prototype that provides transparent, automated care-task accountability and real-time alerts for schedule deviations, thereby strengthening the overall effectiveness of the elder-care monitoring system.

Keywords: Elder-care, pre-trained Activity Recognition Models, pose-based HAR, schedule monitoring, rule-based engine, caregiver accountability

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
HAR	Human Activity Recognition
AAL	Active and Assisted Living
API	Application Programming Interface
REST	Representational State Transfer
RGB	Red Green Blue Image Format
RGB-D	RGB + Depth Data
CCTV	Closed Circuit Television
IoT	Internet of Things
ML	Machine Learning
DB	Database
JSON	JavaScript Object Notation

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INTRODUCTION

Background and Context

The ageing population has increased demand for home-based elder care, where families prefer to keep older adults in familiar home environments supported by hired caregivers. These caregivers are responsible for tasks such as feeding, bathing, mobility assistance, and medication administration, yet family members are often unable to continuously supervise whether these tasks are completed on time and according to the care plan due to work and distance. This can lead to missed meals, inconsistent hygiene, or delayed medication, which negatively impact health and quality of life.

Recent advances in computer vision and deep learning have enabled pre-trained Activity Recognition Models that recognize human actions from video, including activities relevant to elderly care [1] [2]. Skeleton- or pose-based HAR approaches take RGB or RGB-D video, extract human skeletons using pose estimation, and then classify activities using sequences of skeletal joint coordinates, capturing motion over time rather than single images [3] [4]. These methods have been successfully applied to indoor activity monitoring and fall-related applications for older adults [5] [6] . In parallel, facial recognition technology in elderly care has been used to identify residents and caregivers, monitor their movements, and support safety monitoring [7] [8].

However, most existing systems focus on answering **"what activity is happening?"** or **"is this behavior abnormal?"**, rather than **"is the caregiver following the defined care schedule?"**. They rarely integrate explicit, time-bound caregiving schedules or automatically verify whether scheduled tasks such as breakfast feeding, or medication have been completed within the planned time window and by the assigned caregiver. Families and care providers, therefore, lack automated, objective evidence of schedule compliance.

This component addresses this need by designing an **Activity Schedule Monitoring and Rule-Based Decision Engine that** evaluates combined outputs from pre-trained pose-based Activity Recognition Models and face recognition against explicit caregiving schedules in a Secure Elder-Care Monitoring Platform.

In this project, pre-trained skeleton-based HAR and face-recognition modules act as upstream services that translate raw video into high-level "what happened, who did it, and when" events, while the proposed component focuses on schedule-aware reasoning rather than training new vision models.

Literature Review

Skeleton-based human activity recognition has become a prominent approach for recognizing actions in video by analyzing sequences of human skeleton poses extracted from RGB or RGB-D data [9] [4]. Research shows that HAR systems using pose sequences can accurately classify a range of daily activities and fall-related events, making them suitable for assisted living and Active and Assisted Living (AAL) scenarios [6] [1]. Studies on skeleton-based human activity classification emphasize feature extraction from joint pairs and temporal modelling to improve recognition accuracy in home environments [3] [5].

In the context of elderly care, HAR algorithms using skeleton data from RGB-D sensors have been proposed to monitor aged people in home environments, recognizing activities relevant to Ambient Assisted Living (AAL) and supporting ageing in place [6] [7]. Deep-learning-based HAR on 3D skeletons has also been applied for human–machine interaction and monitoring purposes [1]. Separately, facial recognition has been implemented in elderly care facilities to identify residents, differentiate them from staff and visitors, and detect signs of distress or discomfort through facial analysis, enabling timely intervention by caregivers [8].

Despite these advances in activity and identity recognition, there is limited research that combines **pre-trained Activity Recognition Model outputs and facial identity** with a structured **schedule engine** that checks whether caregivers follow predefined care routines. Existing systems usually lack an explicit scheduling layer and rule-based logic to classify each scheduled task as on time, delayed, missed, or unauthorized based on recognized actions and caregiver identity [10] [11]. This motivates the proposed schedule monitoring module.

Research Gap

Recent research in **Human Activity Recognition** has shown that skeleton-based and pose-based activity recognition techniques can effectively detect daily human activities such as walking, sitting, eating [9] [4] [1]. These approaches are widely used in **Ambient Assisted Living** systems to monitor elderly individuals and ensure their safety in home environments [6] [7]. Additionally, facial recognition technologies have been applied in elderly care systems to identify residents and caregivers, improving security and enabling personalized monitoring [8].

Although these technologies successfully detect activities, identities, and abnormal behaviors, several important limitations remain in current elderly monitoring systems.

Key Limitations in Existing Systems

Lack of structured care schedule modelling

Most existing systems focus primarily on detecting activities or anomalies, such as falls or unusual movements, using activity recognition techniques [4] [1]. However, elderly care in real-world environments typically follows structured daily routines. Activities such as waking up, eating meals, taking medication, and bathing are usually performed according to predefined schedules.
Current monitoring systems rarely represent these routines as structured care tasks that include:

- Defined **time windows** (e.g., breakfast between 8:00–8:30 AM)
- Assigned **caregivers responsible for the task**
- **Task priorities** for critical activities such as medication

While some research discusses general elderly care management in AAL environments [12] [6], structured task-schedule modelling is not commonly addressed in activity recognition systems.

Absence of schedule-aware decision mechanisms

Most elderly monitoring systems generate **raw detection outputs**, such as:

- "Eating detected"
- "Walking detected"

These outputs indicate that an activity occurred but do not interpret whether the activity occurred within an expected care schedule. As a result, these systems cannot determine whether a care routine has been followed correctly.
Existing monitoring platforms and caregiver support tools primarily focus on recording or reminding care routines rather than automatically evaluating them using intelligent monitoring systems [10] [11].
Therefore, current systems lack mechanisms to classify detected activities as:

- **ON_TIME**
- **DELAYED**
- **MISSED**
- **UNAUTHORIZED**

This limits the usefulness of activity recognition results for evaluating real caregiving performance.

Limited caregiver accountability

Most elderly monitoring solutions prioritize patient safety and anomaly detection, such as fall detection or unusual activity monitoring [6] [7]. rather than verifying adherence to structured care schedules. Although facial recognition and tracking methods can identify individuals, they are rarely coupled with explicit task schedules and caregiver assignments to evaluate whether care duties are performed as planned. [8].

As a result, existing systems generally do not monitor:

- Whether each assigned caregiver performs their scheduled tasks within the correct time window defined in the care plan (for example, breakfast between 08:00–08:30 or medication at 10:00).
- Whether the caregiver is present in the required room or zone (such as bedroom, bathroom, or kitchen) when carrying out the scheduled task.
- Whether the detected activity can be linked to completion of a specific scheduled care task for a particular elder, with an explicit status such as ON_TIME, DELAYED, MISSED, or UNAUTHORIZED.

This creates a gap in ensuring **caregiver accountability and routine compliance** in elderly care environments.

Identified Research Gap

Based on these limitations, there is a need for an intelligent monitoring system that not only detects activities and identities but also **interprets these events within the context of predefined care schedules**.

Proposed Solution

This component addresses the identified gap by introducing a schedule-aware, rule-based decision engine that integrates outputs from pre-trained pose-based activity recognition models and face-recognition systems with structured elderly care schedules. The system focuses on a set of basic daily activities that are commonly supported by existing skeleton-based HAR models, including:

- Walking
- Sitting/resting
- Eating meals
- Drinking (hydration)
- Going to bed / lying down
- Getting up from bed

The proposed component performs the following functions:

- Receives activity recognition outputs for the above activities as discrete events with timestamps and confidence scores.
- Receives identity verification results for caregivers and elders from a face-recognition module.
- Compares these fused events against predefined care schedules that specify time windows, locations (zones), and assigned caregivers for each task.
- Evaluates task execution using deterministic, rule-based reasoning over time, identity, and zone.

Based on this analysis, the system automatically classifies care tasks into the following categories.

- ON_TIME – Task-related activity detected within the scheduled time window and performed by the assigned caregiver in the expected zone.
- DELAYED – Task-related activity detected after the scheduled time window but within the same schedule period.
- MISSED – No matching task-related activity detected by the end of the allowed time window (including any grace period).
- UNAUTHORIZED – Task-related activity detected, but performed by an unidentified or unauthorized individual.

OBJECTIVE

Main Objective

To develop an Activity Schedule Monitoring and Rule-Based Decision Engine that uses outputs from **Pre-trained, external** pose-based Activity Recognition Models and caregiver identity events to automatically verify whether scheduled caregiving tasks for elderly individuals are completed on time, correctly, and by authorized caregivers.

Specific Objectives

- Design a **care schedule data model** representing caregiving tasks, time windows, assigned caregivers, zones, and priority levels.
- Implement backend services that allow administrators to **create, update, and retrieve** schedules for each elderly person.
- Integrate **pre-trained pose-based Activity Recognition Models** (video-based HAR) via the main AI subsystem to provide activity events for actions such as eating, bathing, walking, and sitting, based on skeletal motion sequences.
- Ensure that the person identity is obtained from a face-recognition module, and that both activity and identity events are fused into a unified event stream before evaluation by the schedule monitoring engine.
- Develop a **rule-based decision engine** that compares fused activity–identity events against schedule entries to classify tasks as ON_TIME, DELAYED, MISSED, or UNAUTHORIZED.
- Generate **deviation events and alerts** when tasks are not detected within the scheduled window or occur outside allowed conditions, and expose them to the main dashboard and notification subsystem.

- Evaluate the module using scenario-based test data and simulated event streams from pre-trained Activity Recognition Models, measuring task detection accuracy and schedule-compliance classification quality.

METHODOLOGY

Research Approach

A **design science research** approach is adopted:

- **Problem identification:** absence of schedule-aware caregiver accountability in current AI-based elder-care systems.
- **Artefact design:** development of the schedule data model and rule-based decision engine.
- **Implementation:** realization of the artefact as a backend component integrated with pre-trained Activity Recognition Models and face recognition.
- **Evaluation:** scenario-based experiments with synthetic schedules and HAR-derived events.

Technical Methods and Algorithms

- **Pre-trained pose-based Activity Recognition Models:**

The main AI subsystem integrates video-based HAR models trained on public skeleton-activity datasets (e.g., NTU RGB+D or similar), which are treated as external, pre-trained services in this project. These models take sequences of detected skeleton poses as input and output activity labels (e.g., EATING, WALKING, BATHING) with timestamps and confidence scores. Pose detection extracts joint coordinates for each person in each frame; consecutive frames are then fed into the HAR model to produce activity events that are forwarded to the schedule engine. This component does not train or fine-tune HAR models; it only consumes their outputs.

In this project, the implementation of camera capture and low-level vision pipelines (e.g., OpenCV, pose estimation, and any TensorFlow/PyTorch-based HAR models) is handled by the main AI subsystem of the platform and not by this component; the schedule engine only consumes the already recognized activity events it exposes.

- **Face recognition and identity:**

A separate face-recognition module identifies caregivers and elders and produces identity events (e.g., person_id, role) from camera streams. A separate face-recognition module identifies caregivers and elders and produces identity events (e.g., person_id, role) from camera streams, again as an external pre-trained service; the proposed component does not modify or train face-recognition models.

- **Event fusion:**

The system fuses identity and activity information by associating skeleton-based activity events with recognized faces in the same scene to form unified events:

`FusedEvent: {elderly_id, caregiver_id, activity_type, timestamp, zone, confidence}`

In practice, this means the vision layer emits discrete FusedEvent records, and the schedule engine operates purely on these records rather than on raw video frames

- **Rule-based decision engine:**

  Deterministic rules compare fused events to schedules, for example:

  - IF `activity_type == scheduled_task` AND `timestamp ∈ [start_time, end_time]` AND `caregiver_id == assigned_caregiver` → `TaskStatus = ON_TIME`.
  - IF `activity_type == scheduled_task` AND `timestamp > end_time + grace` → `TaskStatus = DELAYED`.
  - IF no suitable activity detected by `end_time + grace` → `TaskStatus = MISSED`.
  - IF `activity_type == scheduled_task` AND `caregiver_id != assigned_caregiver` → `TaskStatus = UNAUTHORIZED`.

Tools and Platforms

- **Programming language: Python**

Python is selected as the primary implementation language because it is widely adopted for both backend development and AI/ML integration, and can readily consume outputs from pre-trained HAR and face-recognition services exposed by the main platform. Its ecosystem (e.g., requests, Flask/FastAPI libraries, database drivers) supports rapid development of REST APIs, rule logic, and data access, which is appropriate for a fourth-year research prototype.

- **Backend framework: Flask or FastAPI**

Flask and FastAPI are lightweight Python frameworks designed for building RESTful APIs that the overall system can call to submit fused events and retrieve task-status information. Flask offers a simple structure that is easy to implement within the project timeline, while FastAPI additionally provides higher performance and automatic OpenAPI/Swagger documentation, which is useful for clearly specifying schedule and status interfaces.

- **Database: MongoDB**

MongoDB, a document-oriented NoSQL database, is used to store flexible JSON-like documents representing schedules, fused events, and task-status records. This aligns with the event-centric nature of the component, where data schemas may evolve with additional metadata (for example, anomaly flags or confidence values), and where efficient querying by elderly_id, caregiver_id, task_type, and timestamp is required for dashboards and reports.

- **AI integration: external pre-trained models**

Pose-based Activity Recognition Models and face-recognition modules are treated as external, pre-trained services that expose web APIs returning activity and identity labels. In the proposed work, these services form the AI layer of the platform, while the component described in this proposal focuses on schedule modelling, event fusion, and rule-based decision logic. This approach enables the project to reuse state-of-the-art models from the literature without undertaking new deep-learning training within the component.

Consequently, this component does not implement camera capture code or train deep learning models (e.g., Mediapipe, TensorFlow, or PyTorch networks) itself. It treats these capabilities as upstream services and focuses solely on schedule modelling, event fusion, and rule-based decision logic over the fused events provided by the AI layer.

Technical Methods and Algorithms

- **Face recognition and identity**

A separate face-recognition module, implemented as an external pre-trained service, will identify caregivers and elders from camera streams and generate identity events (e.g., {person_id, role, timestamp}). This project does not design or train face-recognition models; it only **consumes** these identity events together with activity labels from the HAR module for schedule-compliance checking.

- **Data Collection and Ethical Considerations**

For this component, data is represented as abstract events (activity label, elderly_id, caregiver_id, timestamps, zone) rather than raw images or video, which reduces direct privacy risks associated with storing or processing identifiable visual data. Scenario data for testing will be created using synthetic schedules and simulated event streams inspired by publicly described HAR methods and datasets, so no new personal data from real elders or caregivers will be collected during the project.

Validation Strategy and Performance Metrics

- **Scenario-based testing:**

  Define sample care schedules (e.g., Breakfast 08:00–08:30, Bathing 09:00–09:30, Medication 10:00–10:10) and generate fused activity–identity event streams representing correct, delayed, missed, and unauthorized behaviors.

- **Metrics:**
    - Task-status classification accuracy (correct ON_TIME/DELAYED/MISSED/UNAUTHORIZED labels).
    - Precision and recall for MISSED and UNAUTHORIZED detections.
    - Qualitative evaluation of rule outputs for clarity and usefulness to admins.

Data Collection

This component operates entirely on abstracted event data instead of raw images or video to reduce privacy risks.

Event representation: All inputs to the schedule engine are represented as FusedEvent records containing fields such as elderly_id, caregiver_id, activity_type, timestamp, zone, and confidence, rather than storing RGB/RGB-D frames.

- **Synthetic schedules:** A set of realistic care schedules (for example, meals, bathing, mobility support, medication) will be manually defined with explicit time windows, assigned caregivers, and priority levels for different scenario types (on-time, delayed, missed, unauthorized).

- **Simulated event streams**: Using these schedules, synthetic fused event streams will be generated to emulate the outputs of upstream HAR and face-recognition services under normal and abnormal caregiving behavior (correct execution, delays, non-execution, wrong caregiver, out-of-zone activity).

- **No real-person data:** The project will not capture or store any new video, image, or biometric data from real elders or caregivers; instead, it will rely on simulated events inspired by behaviors described in the literature and by typical home-care routines.

Evaluation Plan

The evaluation focuses on how accurately the rule-based engine classifies schedule compliance and generates appropriate alerts when given fused activity–identity event streams.

- **Scenario design:** A suite of test scenarios will be constructed, each consisting of
    (i) a ground-truth care schedule and
    (ii) a labeled timeline of expected outcomes for each task (ON_TIME, DELAYED, MISSED, UNAUTHORIZED).

- **Functional correctness:** For each scenario, engine outputs (task status labels and deviation events) will be compared against the ground-truth labels to compute: task-level accuracy, precision, recall, and F1-score for each status category.

- **Robustness checks:** Additional stress scenarios will introduce noisy or borderline input (for example, events arriving near the boundary of time windows, overlapping tasks, or missing identity fields) to verify that the rules handle edge cases consistently.

- **Performance and scalability:** For larger synthetic datasets, basic runtime metrics (processing time per event batch, database query latency for schedule lookups) will be observed to ensure that the prototype can handle continuous event streams at a practical scale for a home-care deployment.

Where appropriate, summary tables and confusion matrices will be reported to illustrate how well the engine distinguishes between ON_TIME, DELAYED, MISSED, and UNAUTHORIZED task outcomes.

Limitations and Challenges

Although the proposed component aims to strengthen caregiver accountability, several limitations and challenges must be acknowledged.

- **Dependence on upstream models:** The accuracy of schedule-compliance decisions fully depends on the correctness of the upstream pose-based HAR and face-recognition services; misclassifications or missed detections at that layer will directly propagate as incorrect task-status labels.

- **Synthetic evaluation only:** Because the project relies on synthetic schedules and simulated event streams rather than real-world deployments, evaluation may not capture the full variability, noise, and unexpected behaviors present in actual home-care environments.

- **Rule-based rigidity:** The use of deterministic rules makes the decision engine easier to explain but potentially less flexible in ambiguous situations (for example, overlapping activities, partial task completion, or multi-caregiver interactions), which could require more advanced probabilistic or learning-based reasoning in future work.

- **Limited context awareness:** The component focuses on whether a task was performed within a specified time window and by an authorized caregiver; it does not assess qualitative aspects such as care quality, duration adequacy, or elder comfort.

- **Integration and deployment constraints:** Real-world deployment would involve integrating with heterogeneous camera setups, network conditions, and institutional policies, as well as dealing with legal and ethical constraints around continuous monitoring and long-term data retention, which are only partially addressed by the current prototype.

HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the proposed solution and its components

The proposed solution monitors elder-care activities by comparing real-world events against predefined care schedules and then classifying each scheduled task as ON_TIME, DELAYED, MISSED, or UNAUTHORIZED. It operates on fused activity–identity events generated by the AI layer, which transforms raw video into abstract "who did what, where, and when" information instead of exposing frames directly. This architecture follows recent work in elderly monitoring that uses skeleton-based activity recognition and privacy-preserving event streams for backend reasoning.

Inputs

- Fused activity–identity events from the AI layer
    - Output of pose-based Human Activity Recognition (HAR) and face-recognition modules
    - Provides privacy-safe labeled events instead of raw video

Subcomponents

1. **Schedule Management Service**
    - Provides REST APIs to **create, update, and delete caregiving schedules**
    - Stores schedules in **TaskSchedule table**: task_id, elderly_id, caregiver_id, task_type, start/end time, zone, priority

2. **Rule-Based Decision Engine**
    - Consumes fused events and active schedule entries for each elder.
    - Applies deterministic temporal and identity rules to determine task status:
        - ON_TIME – task-related activity detected within [start_time, end_time + grace] by the assigned caregiver in the expected zone.

        - DELAYED – task-related activity detected after end_time + grace but still on the same schedule day.
        - MISSED – no matching activity detected by end_time + grace.
        - UNAUTHORIZED – task-related activity detected but performed by an unassigned or unknown caregiver.

3. **Task Status Store**

    - Saves task results in **TaskStatus table**: task_id, status, timestamp, details
    - Feeds dashboards and reporting services

Outputs

- **Admin / Family Dashboard**
    - Real-time view of current tasks and their status (ON_TIME, DELAYED, MISSED, UNAUTHORIZED).
    - Historical timelines and summaries for each elder and caregiver
- **Notification Subsystem**
    - Deviation alerts for MISSED, significantly DELAYED, or UNAUTHORIZED tasks.
    - Configurable channels and severity levels depending on task priority (for example, medication)

Additional Design Highlights

The component focuses solely on schedule modeling, event fusion, and rule-based reasoning, while treating pose-based HAR and face recognition as external, pre-trained services.

By processing abstract fused events instead of raw video, the backend:

- Reduces network bandwidth requirements
- Improves privacy and data security
- Simplifies scaling across multiple homes or care facilities

Utilizes validated datasets (Kinetics-400, UCF101, COCO Keypoints, MPII Human Pose, LFW) and pre-trained models (FaceNet, ArcFace, OpenPose, HRNet, ST-GCN, TSN) to ensure high accuracy and reliability while minimizing training effort.

System diagram of components

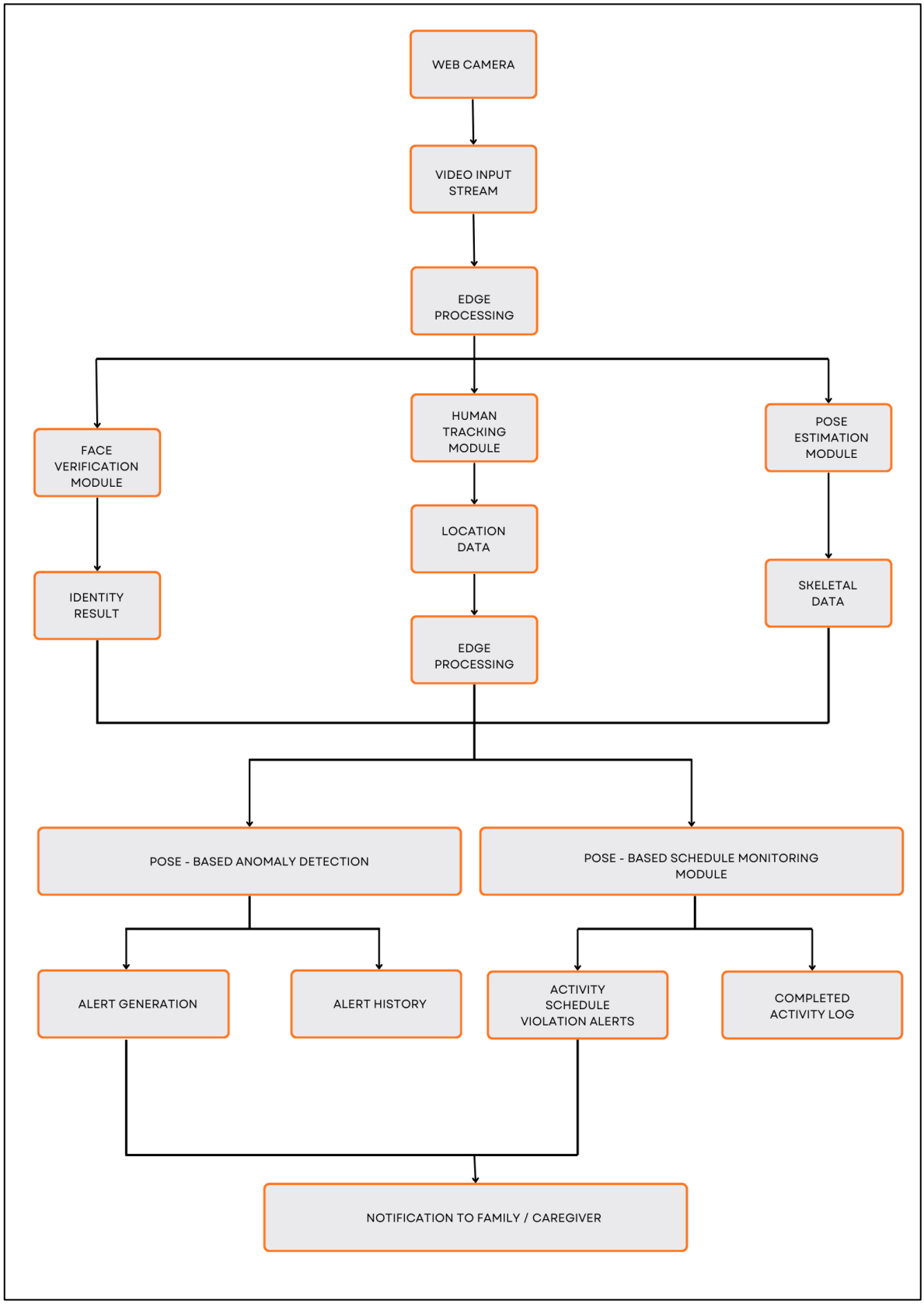


Figure 1: High-Level System Architecture of the Elder Care Monitoring Platform

High-Level System Overview

The overall system combines edge-based video processing, AI modules, and backend microservices to deliver real-time, privacy-preserving elder-care monitoring. Edge devices perform low-level computer-vision tasks, while the cloud/backend focuses on schedule compliance, storage, and visualization. This division of responsibilities reflects common edge–cloud patterns in recent IoT and smart-home monitoring frameworks.

Edge-Based Video Preprocessing and AI

- Captures video from webcams or IP cameras installed in relevant rooms (e.g., bedroom, living room, kitchen).
- Performs frame resizing, noise reduction, motion detection, and frame sampling to lower latency and bandwidth.
- Runs core AI modules locally on an edge device (e.g., Raspberry Pi), including:
    - Face Recognition: Identifies elders and caregivers; emits identity events {person_id, role, timestamp, zone}.
    - Human Tracking: Tracks individuals across frames and rooms, mapping movements into semantic zones (living room, bedroom, kitchen, entrance).
    - Pose Estimation & HAR: Extracts skeletal key points and classifies activities such as eating, walking, bathing, sitting, or falling using pre-trained skeleton-based models.

Context Fusion Layer (Edge or Near-Edge)

- Combines identity, location, and activity information into unified fused events that describe “who did what, where, and when.”
- Aligns multiple streams (face IDs, skeleton tracks, zone labels) in time to generate records like:
    - FusedEvent = {elderly_id, caregiver_id, activity_type, timestamp, zone, confidence}.
- Pushes only these abstract events to the backend via secure APIs, keeping raw video local on the edge wherever possible.

Safety & Schedule Monitoring

- **Anomaly Detection Service (separate component in the full system)**
  - Consumes relevant activity and pose features to detect falls, fainting, or prolonged inactivity.
  - Triggers immediate safety alerts to caregivers and family members when critical anomalies are detected.

- **Activity Schedule Monitoring Engine (your component)**
  - Receives fused events from the AI / fusion layer.
  - Looks up matching schedule entries from the TaskSchedule collection.
  - Applies rule-based reasoning to classify each scheduled task as ON_TIME, DELAYED, MISSED, or UNAUTHORIZED.
  - Emits deviation events to the notification subsystem and stores results in the TaskStatus collection.

Backend & Deployment Architecture

- The schedule-monitoring engine runs as a stateless microservice in the backend, typically behind an API gateway.
- MongoDB stores schedules, fused event logs (optionally), and task-status documents in dedicated collections, enabling efficient queries by elderly_id, caregiver_id, task_type, and time range.
- Edge devices maintain local processing and can continue basic operation (e.g., buffering events) even during intermittent connectivity, syncing with the backend when links are restored.

Key Benefits

- **Enhanced Elder Safety**
  - Real-time anomaly detection from pose and activity data supports rapid intervention for falls or prolonged inactivity.
- **Caregiver Accountability**
  - The schedule engine translates raw activity detections into explicit compliance labels and time-stamped records showing who performed each task and when, supporting audits and family discussions.
- **Privacy-Preserving Design**
  - Raw video stays at the edge; the backend processes only abstract events, reducing privacy risks and regulatory burdens often raised in smart-home elderly monitoring deployments.
- **Efficient and Scalable Operation**
  - Edge processing reduces bandwidth and cloud compute load, while backend microservices and document-based storage make it easier to scale to multiple homes or small facilities.

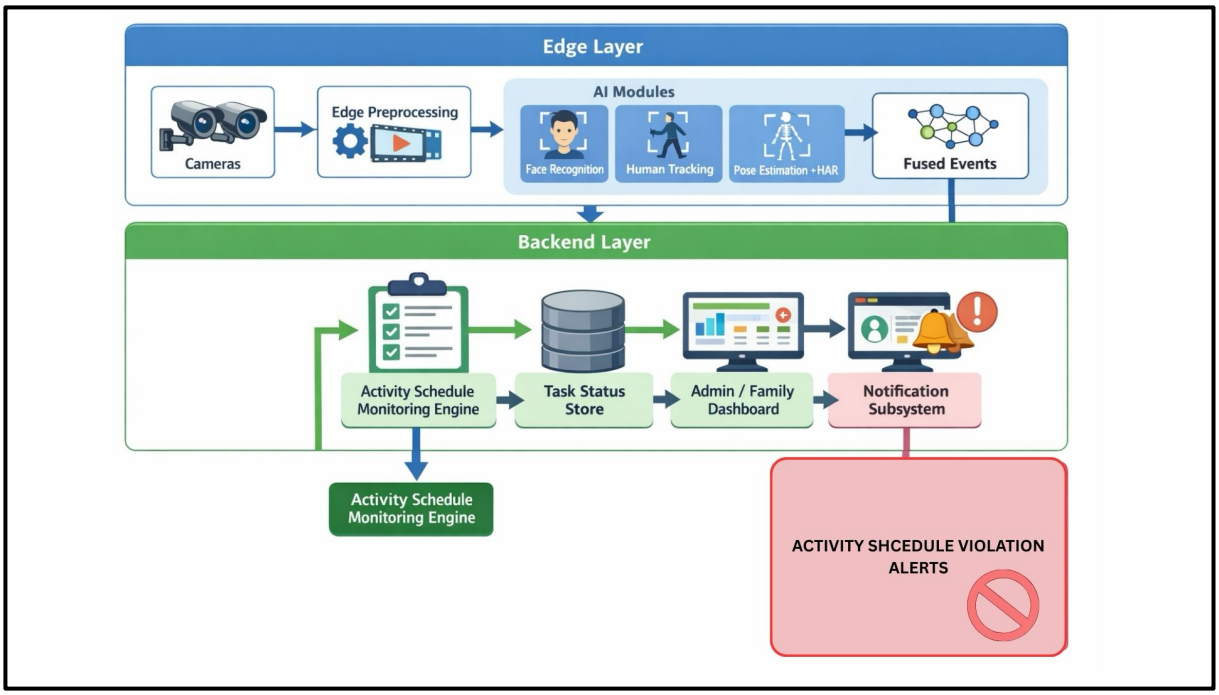


Figure 2:Pose-Based Schedule Monitoring – System Architecture

USER REQUIREMENTS

Stakeholder Identification

- **Admin / Family Member:** Configures schedules, views compliance and receives alerts.
- **Caregiver:** Their actions are assessed against the schedule (indirect stakeholder).
- **System Modules:** Pre-trained Activity Recognition Models, face recognition, anomaly detection.

Functional Requirements

- FR1: The system shall allow administrators to create, modify, and delete caregiving schedules specifying task type, time window, responsible caregiver, zone, and priority.
- FR2: The system shall receive fused activity–identity events produced from pre-trained pose-based Activity Recognition Models and face-recognition modules.
- FR3: The system shall map fused events to scheduled tasks using activity type, time window, and zone.
- FR4: The system shall classify each scheduled task as ON_TIME, DELAYED, MISSED, or UNAUTHORIZED using a rule-based decision engine.

- FR5: The system shall generate deviation alerts for MISSED, significantly DELAYED, or UNAUTHORIZED tasks and provide these to the notification subsystem.
- FR6: The system shall provide APIs for retrieving current and historical task-status information for use in the admin dashboard.

Non-Functional Requirements

- **Performance:** Status evaluation should typically complete within a few seconds after events are received.
- **Reliability:** High accuracy in schedule compliance classification, with low false positives/negatives.
- **Security:** Access to schedules and statuses must be restricted to authenticated, authorized users.
- **Usability:** Task status and history must be easy for non-technical users to interpret.

COMMERCIALIZATION PLAN

- **Target market and customer persona**

The primary target market comprises home-based elder-care settings where families employ caregivers, and small to medium elder-care facilities that either already use, or are willing to adopt, camera-based monitoring.

For this component, the most relevant customer personas are:

- **Remote family decision-maker (home-based elder care)**

An adult child or close relative who lives away from the elderly parent and has arranged a hired caregiver based on a written or informal care plan. They already have (or are willing to adopt) a camera-based monitoring system but do not have time to watch live video or recordings. What they want is an objective, web application-based view that shows whether key scheduled tasks such as meals, bathing, and medication were completed on time and by the assigned caregiver, without having to interpret raw AI outputs or video streams.

- **Care-agency or small facility manager (professional caregiver provider)**

A manager at a small elder-care agency or residential home who employs multiple care givers and is responsible for ensuring that agreed care routines are followed for each resident. They need reliable, automatically generated records of which caregiver performed which task, at what time, and whether it was on time, delayed, missed, or unauthorized, in order to support internal quality assurance, respond to family concerns, and provide documentation during audits or complaints.

These personas align with the background problem described in the introduction (families unable to supervise caregivers continuously and a lack of automated evidence of schedule compliance) and are the primary expected users of the task-status and alert information produced by the proposed Activity Schedule Monitoring and Rule-Based Decision Engine.

- **Value proposition**

The proposed Activity Schedule Monitoring and Rule-Based Decision Engine converts pose-based activity labels and face-based identity outputs into explicit schedule-compliance labels (ON_TIME, DELAYED, MISSED, UNAUTHORIZED) for each caregiving task. This provides families and care organizations with an explainable, time-stamped care log showing who did what, when, and whether it matched the agreed schedule going beyond conventional fall detection or simple presence monitoring and thereby improving transparency, trust, and safety in elder care.

- **Revenue model**

Commercialization is envisaged as a subscription-based add-on within an elder-care monitoring platform. Customers would pay a recurring fee (for example, per monitored elder or per facility) for access to schedule monitoring, real-time alerts, and compliance dashboards, with higher-tier subscriptions offering advanced analytics such as monthly compliance reports, caregiver performance summaries, and exportable documentation for regulators or insurers.

- **Cost estimation**

Key cost elements include software development and maintenance of the schedule engine, database, and APIs, as well as integration work with existing pose-based activity recognition and face-recognition services. Operational costs arise from hosting the backend microservice and MongoDB database, storing event and task-status logs, and providing monitoring and customer support; these costs are moderated because the component reuses pre-trained AI models and operates on abstract events rather than raw video, avoiding expensive training workloads.

- **Pricing strategy**

A tiered pricing strategy is appropriate, with a basic plan aimed at single-household use (one elder, limited history, core alerts and schedule view) and higher tiers tailored to agencies or facilities (multiple elders and caregivers, longer history retention, detailed compliance analytics and report export). Pricing can be benchmarked against existing elder-care monitoring and caregiver management subscriptions and justified by the additional value of objective schedule-compliance assessment, not just incident or fall detection.

- **Competitive advantage**

Existing commercial solutions typically focus on fall detection, SOS alarms, or simple check-in/out and location tracking, with limited support for automatically checking whether detailed care schedules are followed. The proposed component offers a competitive advantage by combining pre-trained pose-based activity recognition and face-recognition outputs with a rule-based schedule engine, generating clear ON_TIME / DELAYED / MISSED / UNAUTHORIZED labels and an auditable history of care tasks that competitors generally do not provide.

- **Intellectual property considerations**

The pre-trained pose-based HAR and face-recognition models used by the system would remain under their original licenses, but the way this component models schedules, fuses activity and identity events, and applies rule-based logic to derive schedule-compliance outcomes can be protected as proprietary software. Potential intellectual property includes the schedule and status data schemas, the decision rules and conflict-handling strategies, the API design for integrating with external AI services, and the methods used to present caregiver accountability metrics and compliance histories within dashboards.

BUDGET AND JUSTIFICATION

Hardware Costs

Table 1:Hardware Cost Estimation

Item	***Description***	***Estimated Cost (USD)***
Raspberry Pi 4 (4GB) or similar edge device	Edge computing device used to host upstream AI services (pose-based HAR, face recognition) and send fused events to the schedule-monitoring backend.	70
Logitech C270 Webcam	Low-cost webcam for capturing video feeds from the elder's environment for pose and face recognition modules.	12
Generic 1080p USB Webcam	Budget-friendly camera for basic testing of activity and identity detection.	15
MicroSD Card (64GB)	Storage for the edge device OS, model files, and temporary event logs.	12
Power Supply for Edge Device	Required to power the Raspberry Pi or equivalent edge computing hardware.	10
Miscellaneous Accessories	Cables, mounts, and stands for positioning cameras in locations such as living room or kitchen.	10

Software License Costs

Table 2:Software License Costs

Software / Service	***Purpose***	***Cost***
Python	Main programming language for backend and rule engine implementation.	Free
FastAPI or Flask	Backend framework for implementing RESTful APIs.	Free
MongoDB Community Edition	Database for schedules, fused events, and task-status documents.	Free
Open-source HAR / Face APIs	Pre-trained pose-based HAR and face-recognition services used upstream.	Free (open-source / research)

Cloud / Online Services Costs

Table 3:Cloud / Online Services Costs

Service	***Purpose***	***Estimated Cost***
GitHub / GitLab	Online repository for source code, issues, and documentation for the schedule engine.	Free (basic)
Optional Cloud Host (e.g., Render / Railway / small VPS)	Hosting a demo instance of the backend API for testing and evaluation.	5–10 USD / month (estimated)
Optional Cloud Storage (e.g., Google Drive / OneDrive)	Backup of scenario data, logs, and experiment results (non-personal).	0–5 USD (estimated)

Human Effort Cost

Table 4:Human Effort Cost

Task	***Estimated Duration***	***Notes***
Literature review, requirements analysis, and conceptual design	3–4 weeks	Initial understanding of prior work and project scope
Design and implementation of schedule data model, APIs, and rule-based decision engine	6–8 weeks	Core backend development
Integration with fused events from pre-trained HAR and face-recognition services, scenario-based testing and evaluation	4–5 weeks	Ensures realistic testing of the schedule engine with upstream AI modules
Documentation, proposal writing, and preparation of demo and evaluation materials	2–3 weeks	Final deliverables for evaluation and demonstration

WORK BREAKDOWN STRUCTURE (WBS)

Task breakdown per member

Member 1 – Face Identity Verification Component

- T1.1: Literature review on face recognition for elderly-care and home monitoring.
- T1.2: Design of face-identity verification flow (registration, arrival detection, identity events).
- T1.3: Integration of pre-trained face recognition model and camera input.
- T1.4: Implementation of "caregiver arrival" detection and notification logic.
- T1.5: Testing with sample caregivers and scenarios (on-time, late, unknown person).

Member 2 – Geofencing / Tracking Component

- T2.1: Literature review on geometric tracking, room-level localization, and geofencing using video.
- T2.2: Design of zones (home vs. outside, living room, bedroom, etc.) and tracking events.
- T2.3: Integration of human detection and tracking to produce location/zone events.
- T2.4: Implementation of home/away detection and in-room presence monitoring.
- T2.5: Testing with multi-zone scenarios (caregiver leaving house, moving between rooms).

Member 3 – Anomaly Detection Component

- T3.1: Literature review on pose-based anomaly detection (falls, fainting, abnormal inactivity).
- T3.2: Selection and integration of pre-trained anomaly / fall-detection models from the internet.
- T3.3: Design of anomaly-event schema (type, severity, timestamp, location).
- T3.4: Implementation of real-time anomaly detection pipeline and alert generation.
- T3.5: Scenario-based testing (simulated falls, fainting, inactivity) and evaluation.

Member 4 – Schedule Rule-Based Detection Component

- T4.1: Literature review on schedule-aware monitoring and rule engines in elder care.
- T4.2: Design of care schedule data model (wake-up, breakfast, medication, bathing, sleep, etc.).
- T4.3: Implementation of schedule management APIs (create/update schedules per elder).
- T4.4: Integration of fused events (identity + activity + location) from components 1–3.
- T4.5: Implementation of a rule-based engine to detect ON_TIME / EARLY / DELAYED / MISSED / UNAUTHORIZED for tasks (e.g., breakfast at 8:30, wake-up at 8:00).
- T4.6: Generation of deviation alerts and task history storage for owner view.
- T4.7: Scenario-based testing of schedule deviations (late breakfast at 10:00, early wake-up at 7:30, missed tasks).

Group-level shared tasks (all members):

- TG.1: Joint high-level architecture design and data/event schema alignment.
- TG.2: Integration of components into one end-to-end pipeline.
- TG.3: Preparation of demo, documentation, and final presentation.

Timeline alignment

- **Month 1**
  - T1.1, T2.1, T3.1, T4.1: Literature review and problem refinement for each component.
  - TG.1: Group architecture and shared data/event schemas.
  - T4.2: Initial schedule model design (because your engine needs clear schedule structure).
- **Month 2**
  - T1.2–T1.3: Face verification flow and model integration.
  - T2.2–T2.3: Zone design and tracking pipeline.
  - T3.2–T3.3: Selection and integration of anomaly models and event schema.
  - T4.2–T4.3: Finalize schedule model and implement schedule management APIs.
- **Month 3**
  - T1.4: Caregiver-arrival notification logic.
  - T2.4: Home/away and room presence logic.
  - T3.4: Real-time anomaly alerting.
  - T4.4–T4.5: Integrate fused events from components 1–3 and implement rule-based schedule engine (ON_TIME / DELAYED / MISSED / UNAUTHORIZED).
- **Month 4**
  - T1.5, T2.5, T3.5, T4.6–T4.7: Scenario-based testing and evaluation of each component.
  - TG.2: System integration and end-to-end demo (camera → AI → events → schedule engine → owner notifications).
  - TG.3: Final documentation, thesis/proposal completion, and presentation preparation.

Balanced workload distribution

- Each member owns **one major functional component** (face identification, tracking/geofencing, anomaly detection, schedule rule engine), ensuring clear responsibility and depth.
- All members participate in **common tasks** (architecture, integration, documentation), so knowledge is shared and no component is isolated.
- Your workload (Member 4) spans design (schedule model), implementation (APIs + rule engine), and evaluation (schedule deviation scenarios), which balances analysis, coding, and experimentation and matches the importance of schedule monitoring within the overall system.

GANTT CHART

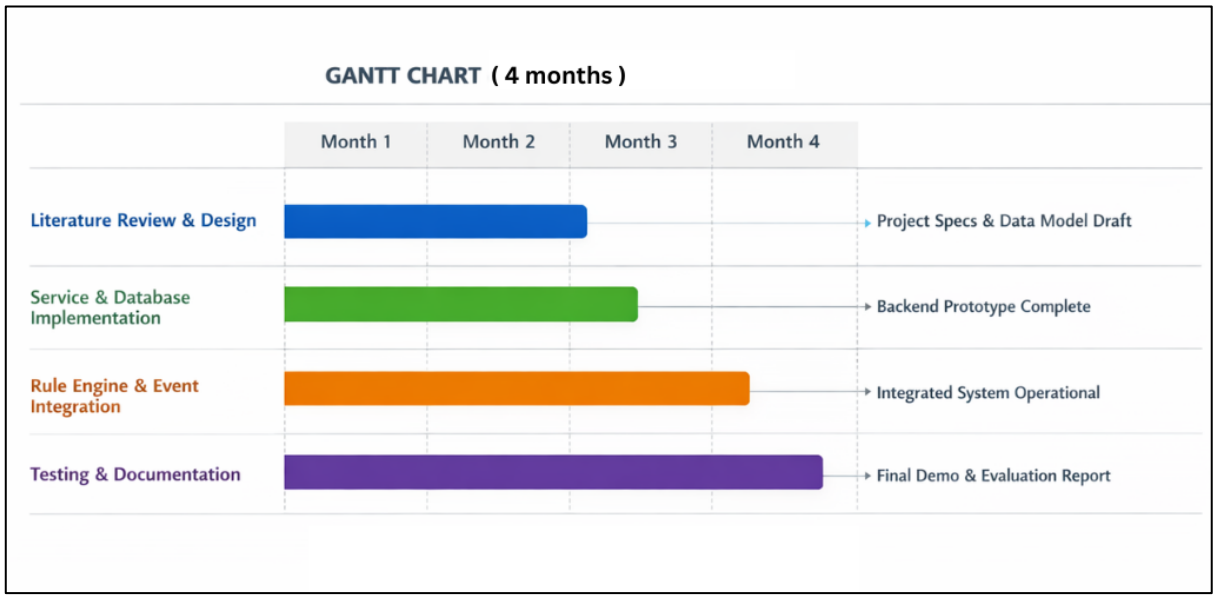


Figure 3:Gantt Chart of the Component

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APPENDICES

A. Survey instruments

This appendix will contain the questionnaires and interview guides used to understand user needs and expectations, including:

- A short survey for family members and elder-care facility staff on how they currently monitor caregivers, what information they would like about arrival, anomalies, and schedule compliance, and their preferred notification methods.
- Optional semi-structured interview questions exploring attitudes towards camera-based monitoring, event-based logs (ON_TIME / DELAYED / MISSED / UNAUTHORIZED), and acceptable levels of automation.

B. Detailed diagrams

System architecture of the Activity Schedule Monitoring and Rule-Based Decision Engine components and their data flows.

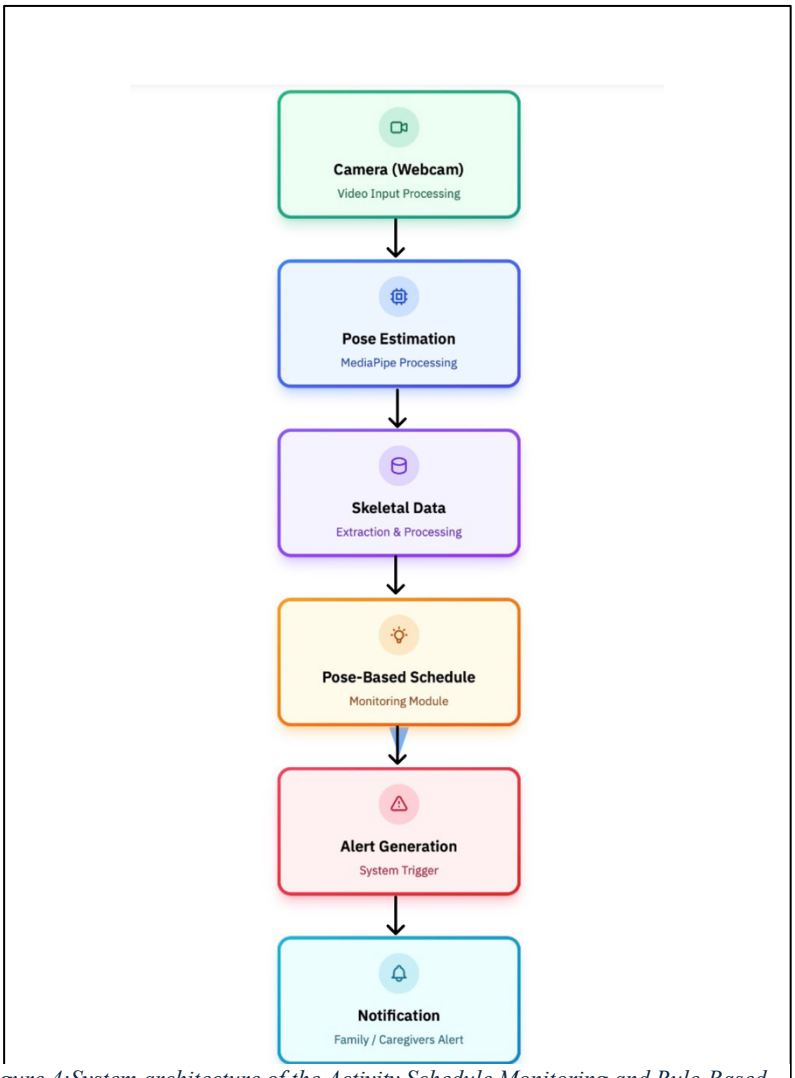


Figure 4:System architecture of the Activity Schedule Monitoring and Rule-Based Decision Engine components

Additional modules in the full system include:

Facial Identity Recognition

- Detects and identifies caregivers to verify their presence and arrival times.
- Ensures that only authorized personnel interact with the elder.

Geofencing / Location Tracking

- Tracks the real-time location of caregivers and elders within the home environment.
- Monitors movements across rooms or predefined zones to support safety and activity monitoring.

Anomaly Detection

- Identifies abnormal behaviors such as falls, prolonged inactivity, or unsafe actions.
- Generates immediate alerts to notify family members or administrators for timely intervention.

C. Preliminary UI mockups

This appendix presents early design concepts for the user interface of the elder care monitoring system. The mockups focus on clarity, usability, and effective communication of schedule adherence to administrators or family members.

Dashboard Overview

- Shows a summary of the elder's daily schedule and task statuses.
- Color-coded indicators for each task: ON_TIME, EARLY, DELAYED, MISSED, UNAUTHORIZED.
- Provides a high-level view of caregiver attendance and overall schedule compliance.

Alerts View

- Displays deviations from the planned schedule for immediate attention.
- Separates scheduled activity alerts (e.g., late meals, missed tasks).
- Includes options to acknowledge alerts or notify family members/administrators.

History / Compliance View

- Presents a historical record of tasks over selected periods.
- Summarizes performance trends for both elders and caregivers.
- Supports easy review of repeated deviations or patterns in schedule adherence

Schedule Management UI

- Allows administrators or caregivers to **insert, update, or delete scheduled tasks**.
- Users can define task type, time window, elder, and assigned caregiver.
- Ensures flexibility in daily or weekly care planning while feeding directly into the rule-based monitoring engine.

Design Principles:

- Focused on readability and quick comprehension.
- Consistent color-coding and clear labels for statuses.
- Intuitive for non-technical users, allowing both monitoring and schedule management in a single platform.

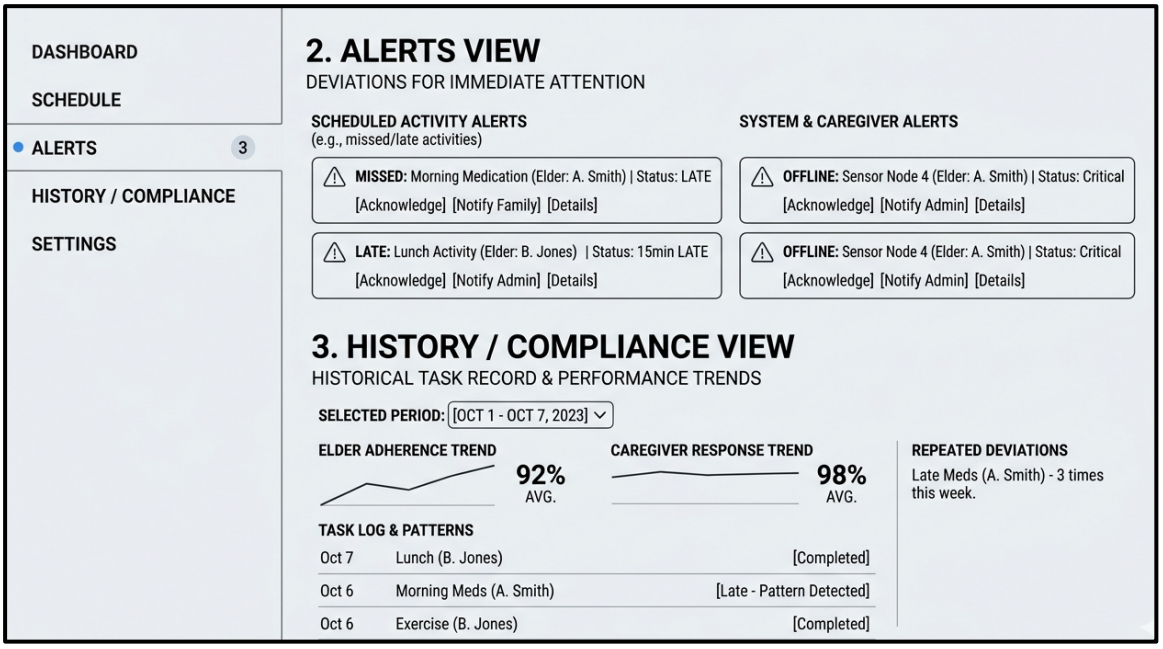


Figure 5:Unified Compliance and Alert Dashboard.

D. Risk analysis

This appendix will summarize the identified technical and ethical risks and proposed mitigation strategies, for example:

Technical Risks:

- Incorrect classification of tasks (ON_TIME, DELAYED, MISSED, UNAUTHORIZED) due to missing or delayed fused events.
- Dependency on upstream modules (pose estimation, face recognition, geofencing) for accurate data.

Ethical / Privacy Risks:

- Perceived privacy concerns from continuous monitoring (even if only abstract events are used).
- Misinterpretation of deviations may cause unnecessary concern for caregivers or family members.

Mitigation Measures:

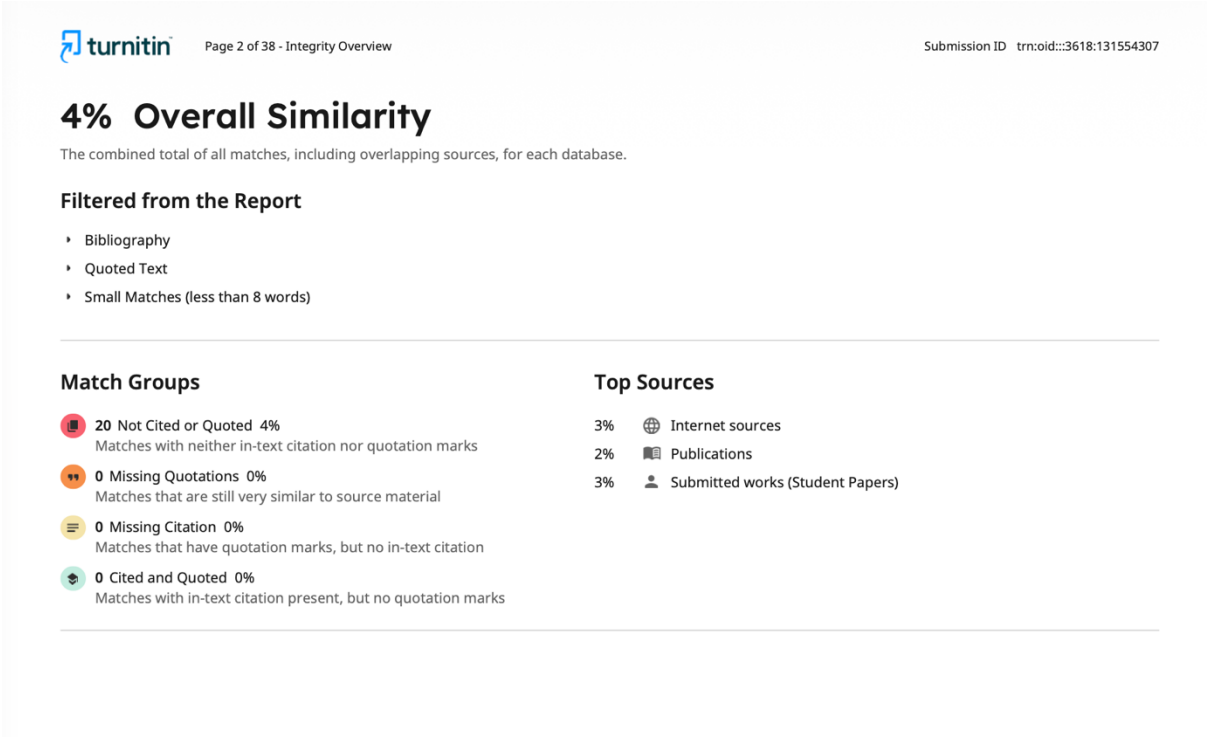
- Use clear rules and time thresholds to reduce false alerts.
- Document assumptions and system limitations.
- Restrict access to task logs and notifications to authorized users only.
- Store only abstract events, not raw video, to protect privacy.

E. Ethical documents

This appendix will compile documents related to ethical compliance, such as:

- Draft participant information sheets and consent forms for elders and caregivers, explaining the purpose of the system, what data are captured (abstract events vs. video), and how alerts and histories will be used.
- Data handling notes describing anonymization or pseudonymization of identifiers in logs used for evaluation, retention periods, and procedures for data deletion on request.
- Any additional institutional ethics forms or declarations required for deployment or evaluation in a real or simulated environment.

F. Turnitin Report



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